Capstone Detailed Outline: Moving from Static to Dynamic Evaluation in AI Language Models

I. Introduction and Background

A. What are Large Language Models (LLMs) and Benchmarks?

- **Understanding LLMs:** Large Language Models (LLMs) are artificial intelligence systems that are increasingly used in our everyday lives, yet we still have very few reliable ways to measure how well they truly understand and process language.
- **What is a benchmark?** A benchmark is essentially a standardized test used by researchers to measure and evaluate the capabilities of different AI models.
- **How benchmarking is usually done:** Most current evaluation methods rely on fixed, pre-written questions that are usually presented in a multiple-choice format, where the AI is shown all possible answers. Because the choices often look very different and there is only one correct option, models can perform well simply by guessing the most statistically likely answer. Consequently, this standard format does not effectively evaluate a model's ability to generate original writing or complete realistic, open-ended tasks.

B. Static Benchmarks vs. Dynamic Benchmarks: The Problem of Data Contamination

- **How static benchmarks work:** A static benchmark (such as MMLU) is a traditional test that features a fixed set of questions that never change over time.
- **Why static benchmarks fall short:** Because these older tests are widely known and unchanging, their questions and answers often end up being included—either accidentally or intentionally—in the text data used to train newer AI models.

- **The "Data Contamination" (Seen Before) problem:** This creates a major flaw where models encounter the test material during their training phase. When the AI takes the benchmark, a high score might just reflect that the model is familiar with the test content from its training, rather than proving it has strong, independent reasoning skills. This makes popular static benchmarks much less informative because they make newer models look artificially smarter than they actually are.

C. The Shift to Dynamic Benchmarks and Their Challenges

- **How dynamic benchmarks work:** To solve the contamination problem, researchers have developed "dynamic" benchmarks (like LiveBench) which frequently update their questions over time (e.g., refreshing questions monthly).
- **Why dynamic testing is better:** Because the test questions are constantly changing, the AI models cannot memorize the answers in advance. This forces the model to encounter entirely new problems, providing a much clearer and more honest picture of how the model performs under real-world conditions.
- **The challenges in dynamic benchmarking:** Creating tests that constantly update is difficult to maintain, and even newer tests that try to be task-based often have parts that remain relatively static. Because of this, even some dynamic tests can eventually face the same data contamination issues over time if they are not fully updated. Furthermore, it is currently rare for researchers to compare these new dynamic tests side-by-side with older static tests under the same, strictly controlled conditions.

D. Purpose and Thesis

- **The goal of this project:** This study will compare how AI models rank when tested on traditional static benchmarks versus dynamic and real-world task-based evaluations.

- **My prediction:** While newer models will likely still outperform older models on the dynamic tests, the massive performance gap seen on static tests will shrink because the newer models can no longer rely on memorized test questions. In certain tasks, an older model may even surprisingly outperform a newer one.

II. The Evaluation Gap in Linguistics

A. Why General Knowledge Tests Fall Short for Linguistics

- **Most standard AI tests focus on general trivia or basic math** rather than deep expertise in a specific field.
- Because of this, **they fail to evaluate how well a model performs on expert-level linguistic analysis**.

B. Some Example Fields of Linguistics Analysis That We Can Test

- **Syntactic Analysis:** Testing if a model can accurately understand the structural rules of language, such as **building or correcting a syntactic tree**.
- **Semantics:** Evaluating how a model handles meaning, specifically its ability to **analyze and resolve ambiguity across different sentence structures** rather than just reading words at face value. (Examples of existing languages understanding benchmarks such as GLUE benchmarks will be part of the analysis.)
- **Metaphorical Analysis:** Assessing the model's capacity to **interpret metaphors and nuanced meaning in context**, ensuring it understands figurative language instead of relying on literal translations.
- **Phonological Analysis:** Examining how well a text-based AI understands the sound systems of language, such as identifying syllable structures, sound patterns, or phonetic transcriptions. Analyze figurative language and metaphors, rather than just interpreting words literally.

III. Comparing the Different Types of AI Benchmark Tasks

A. Static Tests (Example: MMLU)

- Static benchmarks, such as MMLU, are traditional tests using fixed, multiple-choice questions that do not change over time, making them highly vulnerable to data contamination because newer models have often already "seen" and memorized the answers during their initial training.

B. Dynamic Tests (Example: LiveBench)

- Dynamic benchmarks, like LiveBench, combat this memorization problem by frequently updating their test questions, which forces the AI to solve entirely new problems and thereby provides a much more honest and accurate picture of its true reasoning skills.

C. Task-Based Tests (Example: SWE-bench)

- Task-based evaluations, such as SWE-bench, move away from simple multiple-choice questions entirely by giving the AI a practical, open-ended assignment—like reading a real issue, fixing a piece of computer code, and passing automated tests—to see if the model can successfully complete a realistic project from start to finish under real-world constraints.

D. Task-Based Linguistics Analysis (Custom Design)

- Because most existing benchmarks focus on general knowledge and ignore expert-level language skills, I will design a small-scale custom task-based linguistics evaluation to test whether the model can actively perform complex linguistic analyses, such as mapping syntactic structures, resolving semantic ambiguity, interpreting metaphorical meaning in context, and analyzing phonology. **(Given the time constraint, I will try to design a very simple dynamic linguistics benchmark, but if time doesn't allow it, I will just work with the GLUE benchmark)**
- We will also be analyzing the LLM performance using the GLUE language understanding benchmark suite.

IV. How the Study Will Be Conducted (Methodology)

A. Choosing the AI Models

- I will select 3 to 4 different AI models, ensuring a mix of older and newer systems, to compare them under a strictly controlled setup.

B. Running the Three Tests

- Step 1: Give the models a traditional static test (a subset of MMLU) to see their baseline scores.
- Step 2: Give the models a dynamic test (LiveBench) and a task-based test (SWE-bench) to see how they handle fresh, unmemorized problems.
- Step 3: Give the models the linguistic analysis tasks (syntax, metaphor, ambiguity) to see how they handle deep linguistic reasoning.

C. Ensuring Fair Grading

- I will use simple verification methods, such as letting the model try the task three times and taking the majority answer.
- To ensure fairness, each type of models and each type of tests will be treated the same way, the accuracy will be measured based on the average of three.

V. Analyzing the Results

A. Comparing the Scores

- I will map out how the rankings of the AI models change when they move from the static test to the dynamic, task-based, and linguistic tests.
- I will compare the testing results of these LLM on dynamic benchmark to existing static benchmark performance, I will also reference existing benchmark testing data from other papers and sources.

B. Expected Findings

- The results will likely show that while newer models are generally better, the traditional static tests exaggerate just how much better they are.

- We may even find specific linguistic tasks where an older model surprisingly outperforms a newer one.

VI. Conclusion

A. Summary of the Study

- A brief recap of how dynamic and linguistic evaluations provide a more honest picture of AI capabilities than static tests.

B. Why This Matters for the Field

- For Linguistics: It shows that AI still has room to grow when it comes to deep linguistic understanding, like syntax and metaphor.
- For the Public: It proves that we need better, more updated ways to test AI in order to truly show the capability of AI and to develop AI models that are more accurate and perform better in real-life task completion.

VII. References / Resources

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